USV Labs

Overview

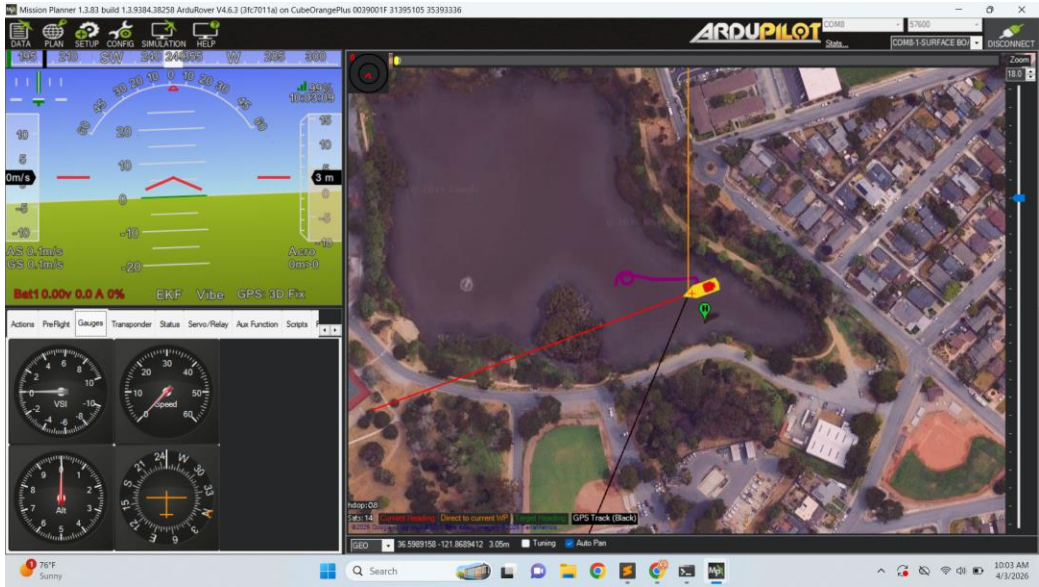
Objectives

Project:
Convert an R/C boat into a USV and navigate a course, consisting of a set of pre-defined waypoints, in minimum time.





Course:
Use the material from class to understand the design and implementation of the ArduPilot control systemto tune the internal feedback control to maximize performance.



Week 1: Make it Work	• Build USVs. • Experiment: Manual control experiment to measure step responses. • Model the system response and find model parameters, based on experiment, to best represent the hardware.
Week 2: Make it Right	• Use ArduPilot documentation and model to simulate closed-loop control of the USV. • Use models to explore the design space. • Experiment: Tune ArduPilot PID gains.
Week 3: Make it Fast	• Competition week! (and write a report)

Lab 1: Measure open-loop response to identify USV model.

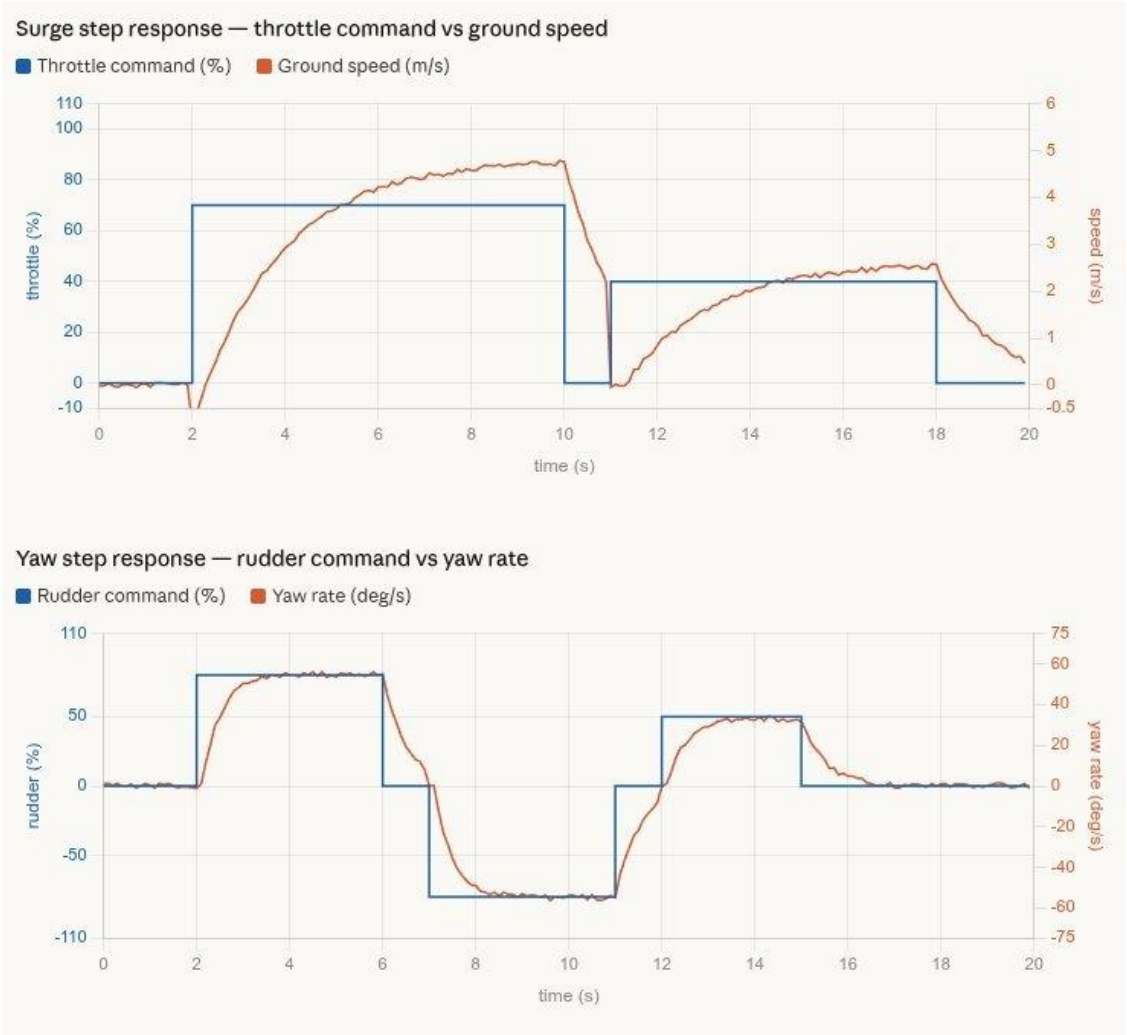
Lab (Mon, Tues?, Wed?):

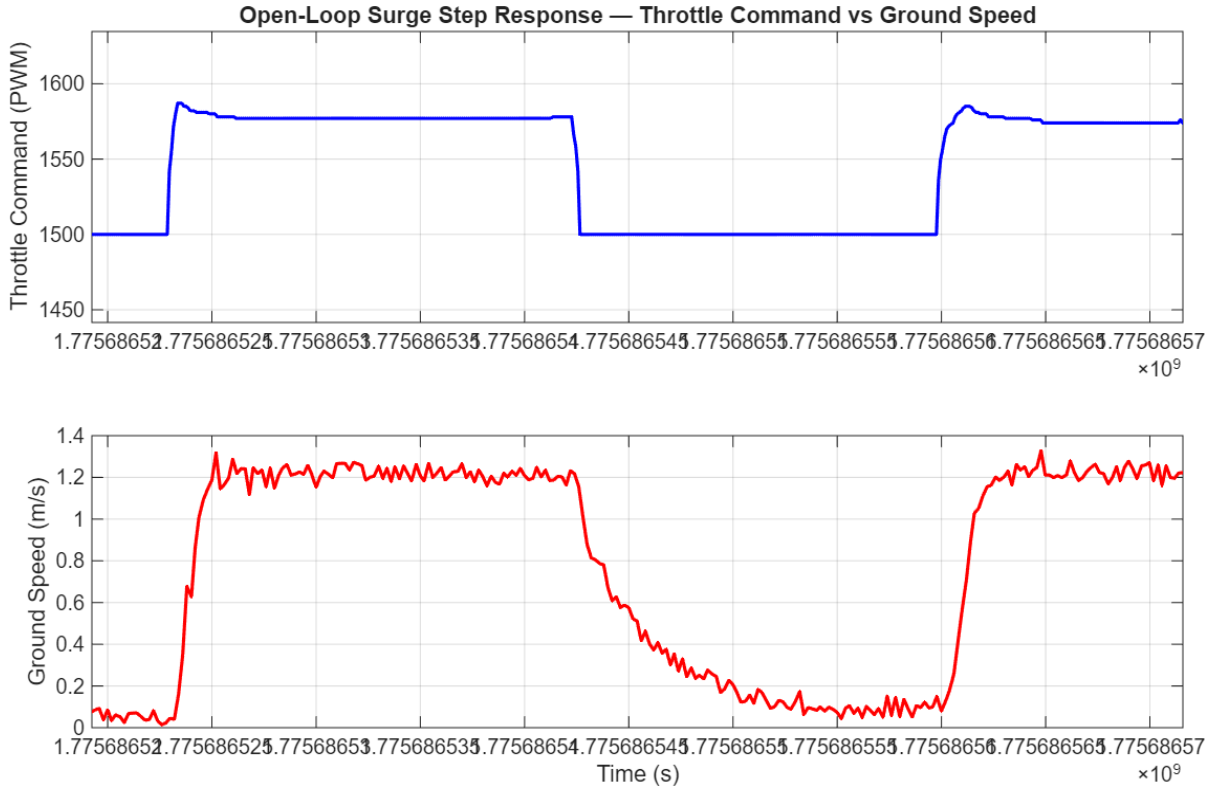
- Build USVs, install Mission Planner, verify R/C control.

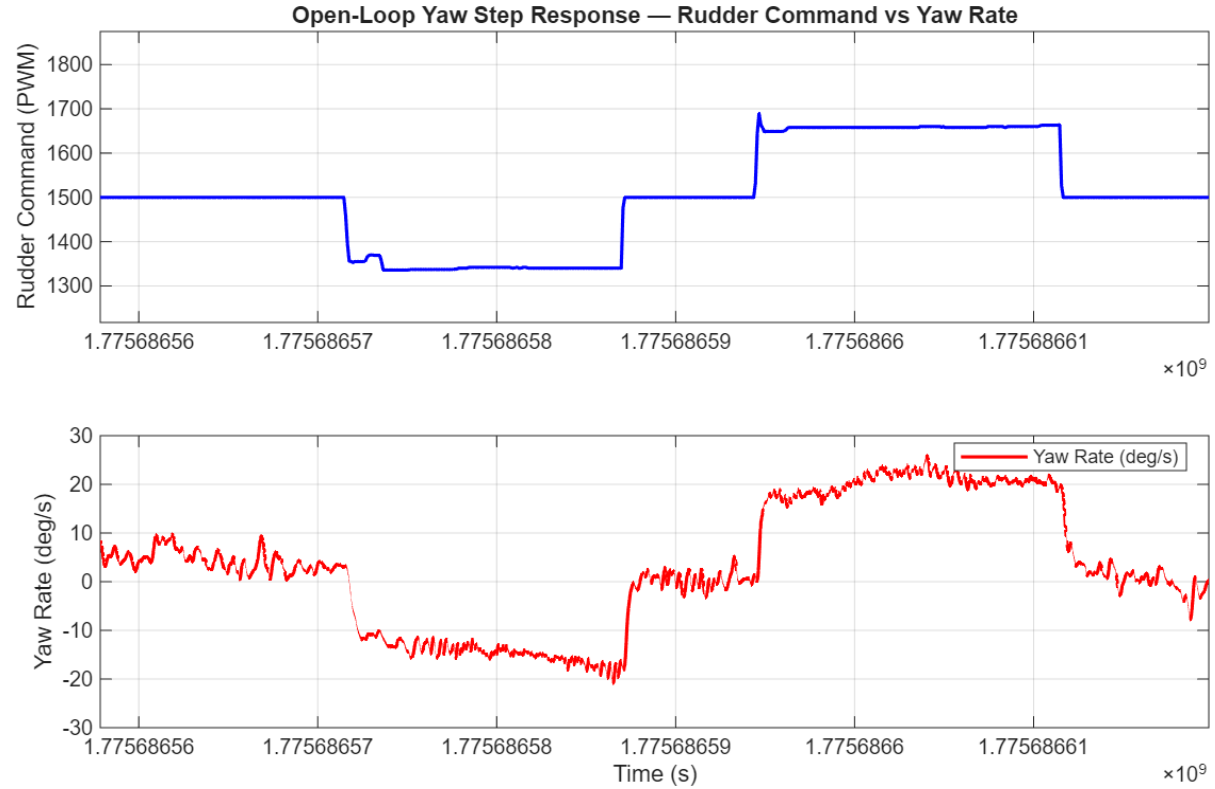
Field (Thurs):

- Log manual step response data.
- Plot step response in speed and yaw-rate.

Write-up due with next week's assignment.




Open-Loop Surge Step Response — Throttle Command vs Ground Speed
Throttle Command (PWM)
1600
1550
1500
1450
1.77568652
1.775686525
1.77568653
1.775686535
1.77568654
1.775686545
1.77568655
1.775686555
1.77568656
1.775686565
1.77568657
×10^9
Ground Speed (m/s)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Time (s)


Open-Loop Yaw Step Response — Rudder Command vs Yaw Rate
Rudder Command (PWM)
1800
1700
1600
1500
1400
1300
1.77568656
1.77568657
1.77568658
1.77568659
1.7756866
1.77568661
×10^9
Yaw Rate (deg/s)
30
20
10
0
-10
-20
-30
Time (s)